# 1. Transportation Systems

Surafel Luleseged (PhD)

Addis Ababa Science and Technology University

March 29, 2026

# Transportation System

Targets:

- Brainstorming from global to Ethiopia's transportation system

# Transportation System

Targets:

- Brainstorming from global to Ethiopia's transportation system
- Highlight challenges which can be discussed on how to deal with them later

# Transportation System

Targets:

- Brainstorming from global to Ethiopia's transportation system
- Highlight challenges which can be discussed on how to deal with them later
- Understanding global trend helps to predict challenges

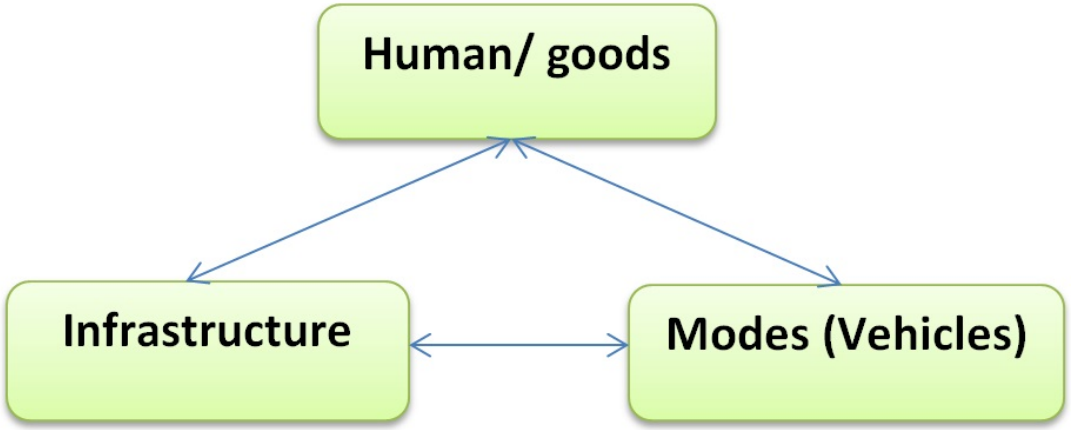
# Transportation System

Targets:

- Brainstorming from global to Ethiopia's transportation system
- Highlight challenges which can be discussed on how to deal with them later
- Understanding global trend helps to predict challenges
- Readiness for challenges from others' experience

# Transportation System

- Movement of people and goods
- Consists:




- Important for country's economic growth

# Transportation Systems

Group discussion on

- Current status of transportation in Ethiopia and especially in Addis Ababa
- What are current and near future advancements
- What are the challanges (including expected)

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally
    - Over 1.4 billion vehicles worldwide

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally
    - Over 1.4 billion vehicles worldwide
    - Global air passengers: 4–4.5 billion annually

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally
    - Over 1.4 billion vehicles worldwide
    - Global air passengers: 4–4.5 billion annually
    - Maritime transport carries 80–90% of global trade volume

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally
    - Over 1.4 billion vehicles worldwide
    - Global air passengers: 4–4.5 billion annually
    - Maritime transport carries 80–90% of global trade volume
  - Advanced infrastructure (e.g., highways, high-speed rail, smart airports)

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally
    - Over 1.4 billion vehicles worldwide
    - Global air passengers: 4–4.5 billion annually
    - Maritime transport carries 80–90% of global trade volume
  - Advanced infrastructure (e.g., highways, high-speed rail, smart airports)
  - Increasing use of AI, IoT, and data science for traffic management

# Global Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Multimodal systems: road, rail, air, and maritime transport are interconnected
    - Road transport carries 75% of inland freight globally
    - Over 1.4 billion vehicles worldwide
    - Global air passengers: 4–4.5 billion annually
    - Maritime transport carries 80–90% of global trade volume
  - Advanced infrastructure (e.g., highways, high-speed rail, smart airports)
  - Increasing use of AI, IoT, and data science for traffic management
  - Strong logistics and supply chain systems

# Global Transportation System Overview

- Current Trends

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport
  - Autonomous vehicles and digital platforms

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport
  - Autonomous vehicles and digital platforms
- Challenges

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport
  - Autonomous vehicles and digital platforms
- Challenges
  - Urban congestion

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport
  - Autonomous vehicles and digital platforms
- Challenges
  - Urban congestion
  - Environmental impact (emissions): transport contributes 23% of global $CO_2$ emissions

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport
  - Autonomous vehicles and digital platforms
- Challenges
  - Urban congestion
  - Environmental impact (emissions): transport contributes 23% of global $CO_2$ emissions
  - Infrastructure maintenance costs

# Global Transportation System Overview

- Current Trends
  - Smart mobility and intelligent transport systems
  - Electrification (electric vehicles), sustainable and green transport
  - Autonomous vehicles and digital platforms
- Challenges
  - Urban congestion
  - Environmental impact (emissions): transport contributes 23% of global $CO_2$ emissions
  - Infrastructure maintenance costs
  - Inequality in access

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade
    - Heavy reliance on ports for international trade

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade
    - Heavy reliance on ports for international trade
  - Air transport growing but relatively costly

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade
    - Heavy reliance on ports for international trade
  - Air transport growing but relatively costly
- Strengths

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade
    - Heavy reliance on ports for international trade
  - Air transport growing but relatively costly
- Strengths
  - Increasing infrastructure investments

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade
    - Heavy reliance on ports for international trade
  - Air transport growing but relatively costly
- Strengths
  - Increasing infrastructure investments
  - Strategic regional transport corridors

# African Transportation System Overview

Globally, transportation systems are highly developed and integrated, supporting economic growth, trade, and mobility.

- Features
  - Road transport dominates (about 70–80%)
    - Total road network: 1 million km
    - Rail network: 90,000–100,000 km
    - Road transport share: 70–80% of freight and passenger movement
    - Transport costs: up to 30–40% higher than global average in some regions
  - Limited rail networks and cross-border connectivity
    - Many countries have limited cross-border rail links
  - Ports are essential for trade
    - Heavy reliance on ports for international trade
  - Air transport growing but relatively costly
- Strengths
  - Increasing infrastructure investments
  - Strategic regional transport corridors
  - Growing demand due to urbanization

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
- Opportunities

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
- Opportunities

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
  - Weak multimodal integration
- Opportunities

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
  - Weak multimodal integration
  - Limited use of digital and data systems
- Opportunities

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
  - Weak multimodal integration
  - Limited use of digital and data systems
- Opportunities
  - Leapfrogging using digital and AI technologies

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
  - Weak multimodal integration
  - Limited use of digital and data systems
- Opportunities
  - Leapfrogging using digital and AI technologies
  - ERegional integration (trade corridors)

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
  - Weak multimodal integration
  - Limited use of digital and data systems
- Opportunities
  - Leapfrogging using digital and AI technologies
  - ERegional integration (trade corridors)
  - Infrastructure maintenance costs

# African Transportation System Overview

- Challenges
  - Poor road quality in rural areas
  - High transport and logistics costs
  - Weak multimodal integration
  - Limited use of digital and data systems
- Opportunities
  - Leapfrogging using digital and AI technologies
  - ERegional integration (trade corridors)
  - Infrastructure maintenance costs
  - Smart urban mobility solutions

# Ethiopian Transportation System Overview

Ethiopia's transport system is rapidly developing, with major investments in infrastructure, but still faces structural challenges.

- Key Components
  - Road transport dominates, accounting for >90% of passenger and freight movement.

# Ethiopian Transportation System Overview

Ethiopia's transport system is rapidly developing, with major investments in infrastructure, but still faces structural challenges.

- Key Components
  - Road transport dominates, accounting for >90% of passenger and freight movement.
  - The road network is expanding (estimated at over 120,000 km).

# Ethiopian Transportation System Overview

Ethiopia's transport system is rapidly developing, with major investments in infrastructure, but still faces structural challenges.

- Key Components
  - Road transport dominates, accounting for >90% of passenger and freight movement.
  - The road network is expanding (estimated at over 120,000 km).
  - Major issues include urban congestion (especially Addis Ababa), road safety, and maintenance gaps.

# Ethiopian Transportation System Overview

- Key Components
  - Rail transport: Addis Ababa–Djibouti Railway ( 750 km, freight and trade) and Addis Ababa Light Rail Transit.

# Ethiopian Transportation System Overview

- Key Components
  - Rail transport: Addis Ababa–Djibouti Railway ( 750 km, freight and trade) and Addis Ababa Light Rail Transit.
  - Rail challenges include limited network coverage and operational integration issues.

# Ethiopian Transportation System Overview

- Key Components
  - Rail transport: Addis Ababa–Djibouti Railway ( 750 km, freight and trade) and Addis Ababa Light Rail Transit.
  - Rail challenges include limited network coverage and operational integration issues.
  - Air transport is led by Ethiopian Airlines, one of Africa's leading aviation hubs.

# Discussion

Group work and discussion

- Past, current status and challanges of public transport in Addis Ababa
- Past, current status and challanges of rail way trasnoprtation in Addis Ababa
- Past, current status and challanges of air trasnoprtation in Addis Ababa